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Carl Zeiss Vision International GmbH
Turnstraße 27
73430 Aalen
Germany

Application: Pioneer for Extended Reality (XR) | Ref. JR_1048257-1

as an Optical System Engineer with a deep passion for optical innovation, I see the development of smart glasses as the perfect convergence of high precision optics and holography display technology, a frontier that is the future and I want to be part of it.

Throughout my career I have specialized in bridging optical prototypes to work in technological applications such as holographic sensors. During my PhD at the Medical University of Innsbruck I built and integrated a full holographic and adaptive optics platform spanning Texas Instruments spatial light modulators, laser sources, and FPGA based real time control. This work focused on holographic phase retrieval and wavefront sensing—technologies that are fundamental to characterizing holographic optical elements (HOE) and integrating them into optical microscope.

My background in optics, holography, spatial light modulation, and interferometry offers a direct bridge to the eye box optimization and holographic projection or waveguide light challenges to tooz smart glasses. I drived projects end to end from idea, component selection through commissioning, and I enjoy the kind of interdisciplinary, fast moving environment that pioneering a new product category demands.

The opportunity to shape the future of mobile computing platforms at a world leading optical company like ZEISS is a compelling one. I would welcome the chance to visit the tooz Extended Reality vision experience.

Kind regards,

Juan David Muñoz Bolaños
PhD Candidate, Medical University of Innsbruck

Juan David Muñoz Bolaños

Optical System Engineer

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Summary

Optical System Engineer specializing in holographic display metrology and wavefront correction optical systems. Expertise in holographic modulation, and bridging optical prototypes to implementation and applications.

Professional Experience

Jan’22 | Present **Med. Univ. Innsbruck** *Innsbruck, Austria*
Optical System Engineer | PhD Researcher

Adaptive Optical Optimization: Developed a Python driven closed loop control system using spatial light modulators to restore signal intensity; applied to active aberration correction in holographic optical elements.

Holographic Wavefront Metrology: Architected high speed holographic phase retrieval and wavefront sensing routines for optical aberration characterization using a C++, Python, and LabVIEW FPGA instrumentation stack.

Holographic Engine Integration: Synchronized high speed display modules, optofluidic modulators, and CMOS sensors to validate holographic concepts; migrated lab setups into robust, portable holographic display prototypes.

Technical Competencies

Optical System Design & Wavefront Control: Knowledge of waveoptics by python simulation and ray optics by ZEMAX OpticStudio, Tolerance Analysis, Holography, Spatial Light Modulators, and interferometric wavefront sensing.

Industrial System Deployment: C++, Python, LabVIEW FPGA for real time instrument synchronization; experience in Gage R&R validation, production floor commissioning, and V Model development cycles.

Machine Vision Software: Halcon (MVTec), Cognex VisionPro, OpenCV; expertise in pattern matching, OCR, and adaptive segmentation for automated inspection and robotic guidance.

Education

Jan’22 | Jun’26 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Adaptive Optics & Precision Metrology

2019 | 2021 **Friedrich Schiller Univ. Jena** *Jena, Germany*
Master of Science in Photonics

2012 | 2018 **University of Cauca** *Popayán, Colombia*
Bachelor of Science in Physics Engineering

Publications & Awards

Awards: Selected for ZEISS SMT and ASLM summer schools, and SPIE Photonics West (San Francisco) Conference. COLFUTURO 2016 to 2018 (best Colombian professionals).

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills & Hobbies

Ownership & Initiative: Lead project focus from ideas to proof of application, holographic wavefront sensor. **Teamwork & Analysis:** Proven collaboration across optics, hardware, and software disciplines with structured root-cause analysis of defects. *Hobbies:* Enjoying the mountains (Biking, Hiking, learning Skiing) • Bachata Dancing • Harmonica • Vision Systems

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
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